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ITCC 15.1

1. Environment Preparation

```
PS C:\Users\Asus\Downloads\Exam_Docker_Practice> docker --version
Docker version 29.2.1, build a5c7197
PS C:\Users\Asus\Downloads\Exam_Docker_Practice>
```

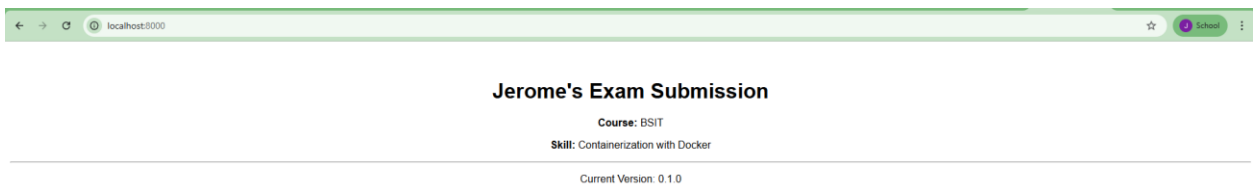
2. Web Application Setup

```
Jerome407 Update README
2650c6f - 3 minutes ago History

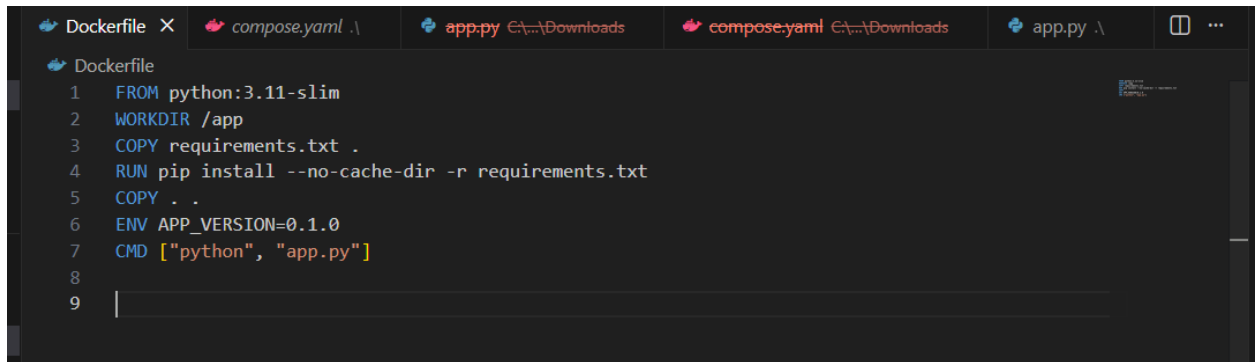
Preview Code Blame 36 lines (16 loc) · 904 Bytes

1 # Jerome's Dockerized CV Web App
2
3 ## Project Overview
4 This is a microservices-based web application with a Flask frontend and a PostgreSQL database backend, built for the Docker configuration exam.
5
6 ## Why flask?
7 I selected the **Flask** framework because it is a lightweight micro-framework that is easy to containerize. Its simplicity allows for rapid development and clear separation of concerns, which is ideal for demonstrating Docker orchestration in this exam
8
9
10
11 ## Architecture
12 - **Web Service:** Flask (Python) running on port 8000.
13 - **Database Service:** PostgreSQL for persistent data storage.
14 - **Orchestration:** Managed via Docker Compose.
15
16 ## How to run
17 1. Clone the repository.
18 2. Run `docker compose up --build`.
19 3. Open `http://localhost:8000` in your browser.
20
21 ## Semantic Versioning
22 This project follows semantic versioning (MAJOR.MINOR.PATCH).
23 Current version: 0.1.0
```

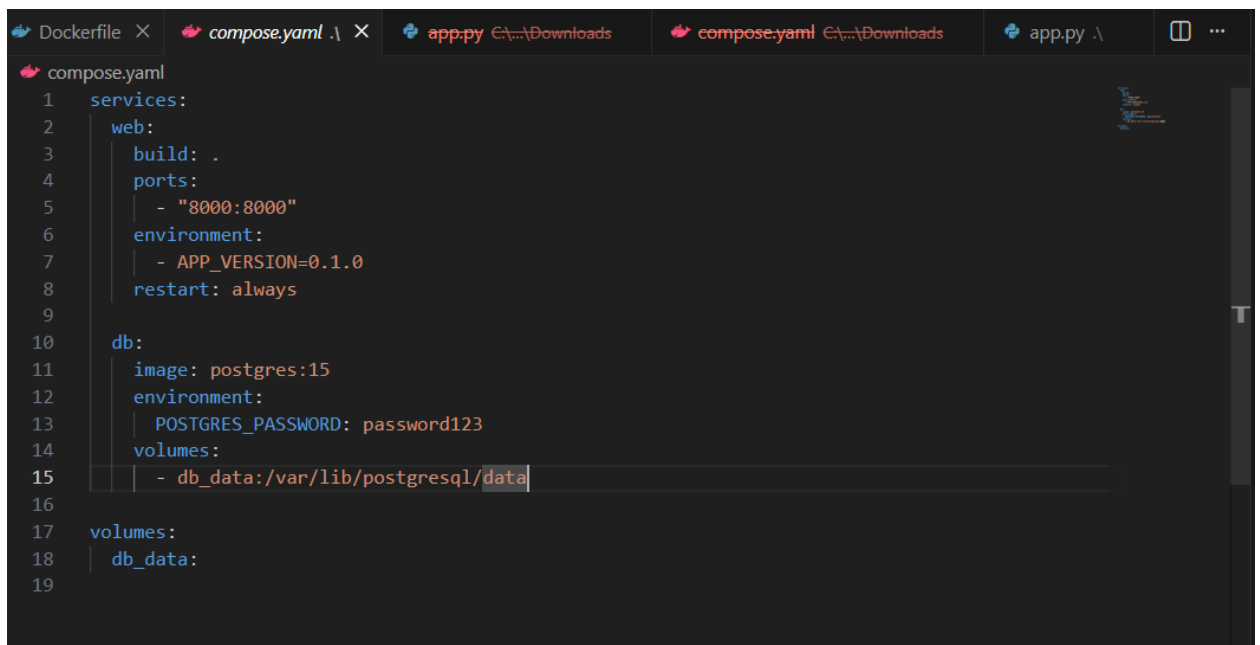
3. Application Configuration



PART 2:

A screenshot of a code editor showing a Dockerfile. The editor has several tabs at the top: 'Dockerfile', 'compose.yaml', 'app.py', 'compose.yaml', and 'app.py'. The 'Dockerfile' tab is active, displaying the following content:

```
1 FROM python:3.11-slim
2 WORKDIR /app
3 COPY requirements.txt .
4 RUN pip install --no-cache-dir -r requirements.txt
5 COPY . .
6 ENV APP_VERSION=0.1.0
7 CMD ["python", "app.py"]
8
9
```

A screenshot of a code editor showing a compose.yaml file. The editor has several tabs at the top: 'Dockerfile', 'compose.yaml', 'app.py', 'compose.yaml', and 'app.py'. The 'compose.yaml' tab is active, displaying the following content:

```
1 services:
2   web:
3     build: .
4     ports:
5       - "8000:8000"
6     environment:
7       - APP_VERSION=0.1.0
8     restart: always
9
10  db:
11    image: postgres:15
12    environment:
13      POSTGRES_PASSWORD: password123
14    volumes:
15      - db_data:/var/lib/postgresql/data
16
17 volumes:
18   db_data:
19
```

4. Semantic Versioning Implementation

I implemented Semantic Versioning (MAJOR.MINOR.PATCH) by setting the initial release to version 0.1.0, which is managed via environment variables in the Docker configuration

5. Advanced Configuration

```

PROBLEMS    OUTPUT    DEBUG CONSOLE    TERMINAL    PORTS

+ FullyQualifiedErrorId : CommandNotFoundException

PS C:\Users\Asus\Downloads\Exam_Docker_Practice> docker --version
Docker version 29.2.1, build a5c7197
PS C:\Users\Asus\Downloads\Exam_Docker_Practice> docker compose up -d
yaml: line 9: mapping values are not allowed in this context
PS C:\Users\Asus\Downloads\Exam_Docker_Practice> docker compose up -d
[+] up 17/17
✓ Image postgres:15-alpine          Pulled
✓ Network exam_docker_practice_app-network Created
✓ Container exam_docker_practice-db-1 Healthy
✓ Container exam_docker_practice-web-1 Recreated
PS C:\Users\Asus\Downloads\Exam_Docker_Practice>

```

Part 3: Build, Deploy, and Testing (25 points)

```

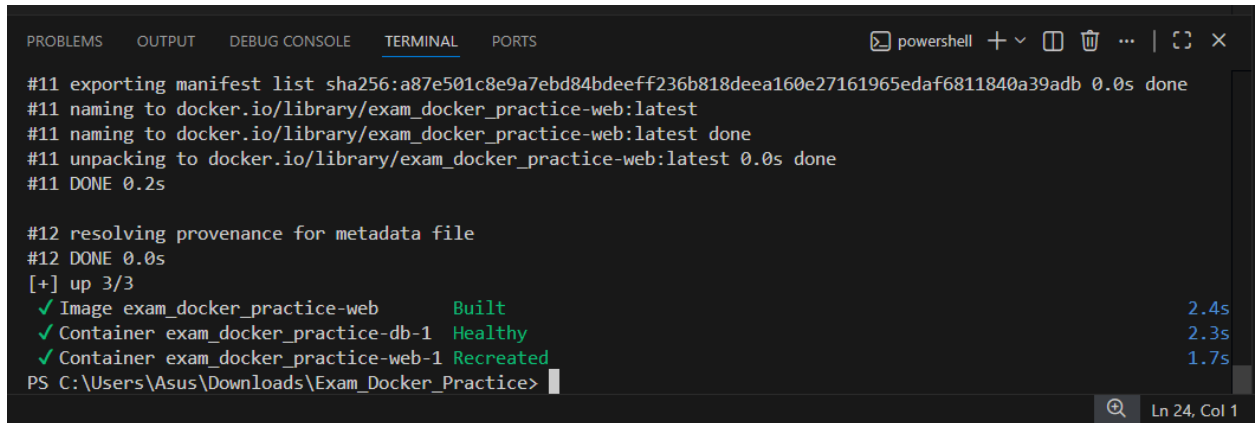
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS
#11 naming to docker.io/library/exam_docker_practice-web:latest done
#11 unpacking to docker.io/library/exam_docker_practice-web:latest 0.0s done
#11 DONE 0.3s

#12 resolving provenance for metadata file
#12 DONE 0.0s

[+] up 4/5
  ✓ Image exam_docker_practice-web Built 3.2s
  ✓ Network exam_docker_practice_app-network Created 0.1s
  ✓ Volume exam_docker_practice_db_data Created 0.0s
  - Container exam_docker_practice-db-1 Waiting 7.8s
  ✓ Container exam_docker_practice-web-1 Created 0.3s

```

2. Integration Testing



```
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS powershell + - [ ] [X] ... | [ ] [X] X
#11 exporting manifest list sha256:a87e501c8e9a7ebd84bdeeff236b818deea160e27161965edaf6811840a39adb 0.0s done
#11 naming to docker.io/library/exam_docker_practice-web:latest
#11 naming to docker.io/library/exam_docker_practice-web:latest done
#11 unpacking to docker.io/library/exam_docker_practice-web:latest 0.0s done
#11 DONE 0.2s

#12 resolving provenance for metadata file
#12 DONE 0.0s
[+] up 3/3
✓ Image exam_docker_practice-web Built 2.4s
✓ Container exam_docker_practice-db-1 Healthy 2.3s
✓ Container exam_docker_practice-web-1 Recreated 1.7s
PS C:\Users\Asus\Downloads\Exam_Docker_Practice>
```

I tested the application lifecycle by performing a version update from 0.1.0 to 0.2.0. I updated the environment variable in the Dockerfile and executed docker compose up --build -d. The process was successful, as Docker recognized the change, rebuilt the application image, and redeployed the container without losing database persistence, demonstrating proper version control and deployment management

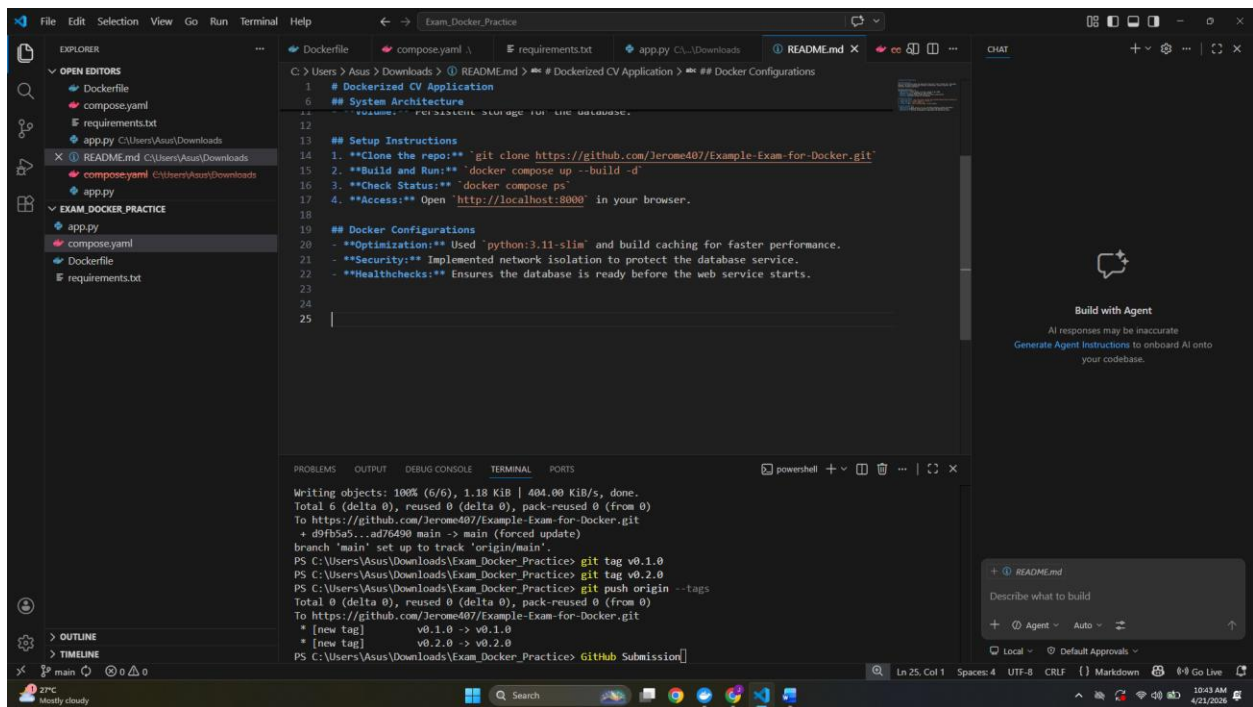
3. Performance and Security

Security Measures:

- **Network Isolation:** Implemented a private bridge network (app-network) to ensure the database service is isolated and only reachable by the web application, not the public internet.
- **Environment Management:** Used Docker Compose environment variables for database credentials to keep sensitive information out of the source code.
- **Performance Optimization:**
 - **Lightweight Base Image:** Used the python:3.11-slim image to reduce the container's footprint, resulting in faster deployment and a smaller attack surface.
- **Build Caching:** Optimized the Dockerfile by copying the requirements.txt separately. This ensures that Docker caches the dependency layer, making subsequent builds much faster.

Part 4: Documentation and Submission

1. Comprehensive Documentation



2. GitHub Submission

**Example-Exam-for-Docker**Public

PinWatch 0

main

1 Branch

2 Tags

Add file

Code

 **Jerome Llyod S Langcao** Final Submission

ad76490 · 8 minutes ago

1 Commit

 Dockerfile	Final Submission	8 minutes ago
 app.py	Final Submission	8 minutes ago
 compose.yaml	Final Submission	8 minutes ago
 requirements.txt	Final Submission	8 minutes ago

README



Add a README

Help people interested in this repository understand your project.

Add a README

Releases

Tags

 Tags

v0.2.0

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...

v0.1.0

8 minutes ago · ad76490 · zip · tar.gz

...